

Social Agents Simulation - Implementation Documentation

1. Overview

This is a multi-agent social simulation built in GAMA Platform that demonstrates how different personality types interact in various social venues. The simulation models realistic social dynamics including movement, face-to-face interactions, long-distance communication (FIPA), and emergent social behaviors.

2. System Architecture

2.1 Global Variables

- **Population Counts:** Controls for 5 agent types (50 total agents by default)
- **Location Counts:** 8 locations across 4 venue types
- **all_locations list:** Global registry to track all locations (fixes parent-child species issue in GAMA)
- **Monitoring Metrics:**
 - `global_happiness`: Average happiness across all agents (0.0-1.0)
 - `total_interactions`, `total_positive_interactions`, `total_negative_interactions`
 - `happiness_history`: Time-series data for trend analysis

2.2 Environment

- **World Shape:** 100x100 square grid
 - **Coordinate System:** Locations spawn within (10,90) range to stay visible
 - **Continuous Execution:** `minimum_cycle_duration <- 0.01` ensures non-stop simulation
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3. Location Species (4 Types)

Base: Location

- **Attributes:**
 - **color:** Visual identification
 - **noise_level:** Environmental characteristic (0.3-1.0)
 - **current_guests:** Dynamic list tracking present agents
- **Aspect:** 5-unit radius circle with border

3.1 Bar

- **Color:** Blue
- **Noise Level:** 0.6-1.0 (noisy)
- **Attracts:** Party People
- **Repels:** Introverts

3.2 Concert

- **Color:** Purple
- **Noise Level:** 0.7-1.0 (very noisy)
- **Special Attribute:** **music_genre** (rock/pop/jazz/electronic)
- **Attracts:** Music Lovers with matching genre preference

3.3 Restaurant

- **Color:** Orange
- **Noise Level:** 0.2-0.5 (quiet)
- **Special Attribute:** **cuisine_type** (italian/asian/vegan/steakhouse)
- **Attracts:** Foodies, Introverts

3.4 SportsVenue

- **Color:** Green
- **Noise Level:** 0.5-0.9 (variable)
- **Special Attribute:** **sport_type** (football/basketball/tennis)
- **Attracts:** Sports Fans with matching preference

4. Guest Species (5 Agent Types)

Base: Guest Species

Skills: [**fipa**, **moving**]

4.1 Personal Traits (3 Required)

Each agent has unique values (0.0-1.0):

1. **Generosity**
 - **Effect:** Probability of sharing resources (drinks, food)
 - **Threshold:** >0.7 triggers sharing behavior
 - **Impact:** Increases recipient's happiness by +0.03
2. **Sociability**
 - **Effect:** Interaction frequency and invitation acceptance
 - **Formula:** $\text{interaction_chance} = 0.2 + (\text{sociability} * 0.3)$
 - **Range:** 20% (low) to 50% (high) interaction probability
 - **Impact:** Determines FIPA message responsiveness
3. **Tolerance**
 - **Effect:** Reduces negative impacts from incompatible situations
 - **Formula:** $\text{noise_penalty} = -0.04 * (1.0 - \text{tolerance})$
 - **Impact:** High tolerance = better coping with mismatches
 - **Friendship:** Higher tolerance = more friends (20% → 50% chance)

4.2 State Variables

- **happiness:** Current emotional state (0.0-1.0, starts at 0.5)
- **target_location:** Where agent is heading
- **current_location:** Where agent currently is
- **has_arrived:** Boolean flag preventing reflex loops
- **time_at_location:** Cycles spent at current venue
- **stay_duration:** Random (30-100 cycles)
- **friends:** List of other agents
- **interacted_this_visit:** Cooldown to prevent spam interactions

4.3 Movement System

Three-State Movement Model:

1. **Moving State** ($\text{has_arrived} = \text{false}$)

gaml

```
reflex move_to_location when: target_location != nil and !has_arrived
```

- Uses **goto** with speed 2.0
 - Checks distance to target
 - When distance < 1.0: Snap to location, set $\text{has_arrived} = \text{true}$
2. **Arrived State** ($\text{has_arrived} = \text{true}$)

gaml

reflex stay_and_interact when: has_arrived and current_location != nil

- Registers in location's `current_guests` list
- Attempts interactions with others
- Increments `time_at_location`
- Leaves when `time_at_location >= stay_duration`

3. Leaving State

gaml

action leave_and_choose_new_location

- Unregisters from `current_guests`
- Resets: `current_location = nil, has_arrived = false`
- Chooses new random target from `all_locations`

4.4 Agent Type Implementations

PartyPerson

- **Color:** Red
- **Trait Ranges:** High sociability (0.7-1.0), High generosity (0.5-1.0)
- **Behavior:**
 - Loves noisy places (`noise_level > 0.6`) → +0.02 happiness
 - Positive with other PartyPerson/MusicLover → +0.03 each
 - Negative with Introverts in noisy places → Introvert gets -0.05
 - **Generosity Effect:** Buys drinks if generosity > 0.7 → recipient +0.03 happiness

Introvert

- **Color:** Light Blue
- **Trait Ranges:** Low sociability (0.1-0.4), Low tolerance (0.2-0.6)
- **Behavior:**
 - Prefers quiet places (`noise_level < 0.5`) → +0.02 happiness
 - Positive with other Introverts/Foodies → +0.03 self, +0.02 other
 - Negative with PartyPerson in any setting → -0.02
 - **Tolerance Effect:**
 - High (>0.6): Only -0.01 with PartyPerson
 - Low (<0.3): -0.03 with PartyPerson
 - Noise penalty scaled by tolerance: $-0.04 * (1.0 - tolerance)$

MusicLover

- **Color:** Magenta
- **Trait Ranges:** High sociability (0.5-0.9), High tolerance (0.6-1.0)
- **Special Attribute:** **favorite_genre** (rock/pop/jazz/electronic)
- **Behavior:**
 - At Concert with matching genre → +0.05 happiness
 - With MusicLover of same genre → +0.04 each
 - With MusicLover of different genre → +0.01
 - With PartyPerson at Concert → +0.02 each

Foodie

- **Color:** Yellow
- **Trait Ranges:** High generosity (0.6-1.0), Moderate sociability (0.4-0.7)
- **Special Attribute:** **diet_preference** (vegetarian/vegan/meat/flexible)
- **Behavior:**
 - At Restaurant with matching cuisine → +0.04 happiness
 - With other Foodie at Restaurant → +0.03 each
 - With Introvert at Restaurant → +0.02 each (quiet conversation)
 - **Generosity Effect:** Shares food if generosity > 0.75 → recipient +0.03 happiness
 - Wrong cuisine type → -0.01

SportsFan

- **Color:** Dark Green
- **Trait Ranges:** High sociability (0.6-0.95), Moderate tolerance (0.5-0.85)
- **Special Attribute:** **favorite_sport** (football/basketball/tennis)
- **Behavior:**
 - At SportsVenue with matching sport → +0.05 happiness
 - With SportsFan of same sport → +0.05 each
 - With PartyPerson at venue → +0.02 each
 - With Introvert at venue → Introvert gets -0.03 (too loud)

5. FIPA Communication System

5.1 Message Types Used

- **inform:** Invitation to join at location
- **agree:** Acceptance of invitation
- **refuse:** Rejection of invitation

5.2 Invitation Sending

gaml

```
reflex send_invitation when: has_arrived and time_at_location > 10
                        and flip(0.1) and length(friends) > 0
...
```

- **Condition**: Must be at location for 10+ cycles, have friends
- **Frequency**: 10% chance per cycle
- **Target**: Random friend at different location
- **Content**: `['invitation', current_location]`

5.3 Multiple Invitation Handling

Decision Algorithm:

```
IF receive 2+ invitations simultaneously:
  IF sociability > 0.7 (High):
    - Prioritize friend invitations
    - If no friends: accept random
  ELSE IF sociability > 0.4 (Moderate):
    - Only accept if from friend
    - Else: reject all
  ELSE (Low):
    - Reject all invitations
```

Responses:

- **Accepted**: Send **agree**, go to sender's location, reject others with **refuse**
- **Rejected Senders**: Receive **refuse** message, get -0.01 happiness penalty

5.4 Message Processing

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```
reflex receive_messages when: !empty(informs)
reflex receive_agreements when: !empty(agreees) // +0.02 happiness
reflex receive_rejections when: !empty(refuses) // -0.01 happiness
```

6. Interaction System

6.1 Interaction Mechanism

- **Frequency**: Based on sociability (20%-50% chance per cycle)
- **Cooldown**: **interacted_this_visit** list prevents repeated interactions

- **Friendship Building:** Based on tolerance (20%-50% chance)
 - Formula: $\text{friendship_chance} = 0.2 + (\text{tolerance} * 0.3)$
 - Mutual: Both agents add each other to **friends** list

6.2 Context-Dependent Interactions

Outcomes vary based on:

1. **Location Type** (Bar vs Restaurant)
2. **Noise Level** (noisy vs quiet)
3. **Agent Type Compatibility**
4. **Personal Traits** (generosity, tolerance)

6.3 Interaction Statistics

- **total_interactions:** Cumulative count
- **total_positive_interactions:** Compatible matches
- **total_negative_interactions:** Incompatible matches
- Used to calculate: $\text{Positive Rate \%} = (\text{positive} / \text{total}) * 100$

7. Monitoring & Visualization

7.1 Global Happiness Metric

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```
reflex update_global_happiness {
  all_guests <- (PartyPerson + Introvert + MusicLover + Foodie + SportsFan)
  global_happiness <- mean(all_guests.happiness)
  happiness_history << global_happiness
}
```

7.2 Charts (2 Required)

Chart 1: Global Happiness Over Time

- Type: Series (line graph)
- Y-axis: 0.0 - 1.0 (happiness scale)
- Data: **global_happiness**, baseline at 0.5
- Purpose: Shows long-term emotional trends

Chart 2: Interaction Quality

- Type: Series (line graph)
- Data: Positive vs Negative interaction counts
- Purpose: Shows social dynamics balance

7.3 Monitors

- Cycle counter
- Global happiness (live value)
- Interaction statistics (total, positive, negative, %)
- Population counts
- Location count

7.4 Visual Display

- **World Boundary:** 100x100 square with black border
 - **Locations:** Semi-transparent colored circles (radius 7)
 - Guest count displayed above each
 - **Agents:** Colored circles (radius 1.5)
 - Color indicates agent type
 - Happiness indicator (small circle above): Red (sad) → Yellow → Green (happy)
-

8. Interesting Conclusions Drawn

8.1 Automatic Analysis (Every 500 Cycles)

gaml

`reflex monitor_stats when: every(50#cycle)`

Logs detailed analysis showing:

Positive Environment (happiness > 0.55):

- Compatible types in appropriate venues thrive
- Example: Introverts in quiet Restaurants

Negative Environment (happiness < 0.45):

- Personality mismatches cause unhappiness
- Example: Introverts forced into noisy Bars

Balanced Environment (0.45-0.55):

- Mixed results, some thrive while others struggle

8.2 Key Findings

Finding 1: Context-Dependent Compatibility

"The same agents can have vastly different experiences based on location characteristics. An Introvert-PartyPerson pair will have negative interactions at a Bar but neutral interactions at a Restaurant."

Finding 2: Personal Traits Matter

"Generosity creates positive ripple effects (+0.03 happiness to recipients), while low tolerance amplifies negative experiences (up to 3x penalty in mismatched settings)."

Finding 3: Social Network Effects

"Agents with high tolerance build larger friend networks (50% vs 20% friendship rate), leading to more FIPA invitations and higher overall mobility."

9. Technical Implementation Details

9.1 Critical Bug Fixes

Problem 1: Parent-Child Species Reference

- `length(Location)` returned 0 despite creating 8 locations
- **Solution:** Global `all_locations` list manually populated during creation

gaml

```
create Bar { all_locations << self; }
```

Problem 2: Infinite Arrival Reflex

- Agents stuck at locations, reflex triggered every cycle
- **Solution:** `has_arrived` boolean flag with proper state management

Problem 3: Agents Not Moving

- Target was `any_location_in(world.shape)` (random point, not venue)
- **Solution:** `target_location <- one_of(all_locations)` (actual venue)

9.2 Performance Optimizations

- Interaction cooldown prevents N^2 complexity
- Chart refresh rates (5/10 cycles) reduce rendering load
- Debug output throttled (every 50-100 cycles)
- Friend list capped at reasonable size via natural turnover

9.3 State Management

Three-state system prevents common GAMA issues:

1. Moving → Arriving → Staying (clean transitions)
2. Proper cleanup on departure
3. No overlapping reflexes due to boolean guards

10. Requirements Checklist

Requirement	Implementation	Status
5+ Guest Types	PartyPerson, Introvert, MusicLover, Foodie, SportsFan	✓
Different Interaction Rules	Each type has unique <code>interact_with()</code> logic	✓
3+ Personal Traits	generosity, sociability, tolerance	✓
Traits Affect Rules	All interactions scale with trait values	✓
2+ Location Types	Bar, Concert, Restaurant, SportsVenue (4 total)	✓
50+ Guests	Default: $12+10+10+10+8 = 50$	✓
Continuous Running	<code>minimum_cycle_duration</code> , infinite loop	✓
FIPA Long-Distance	Invitations across locations with multi-message handling	✓
1+ Global Monitor	<code>global_happiness</code> tracked continuously	✓
1+ Informative Graph	2 charts: Happiness + Interactions	✓
1+ Conclusion	Context-dependent compatibility + trait effects	✓

Expected Behavior:

- Agents continuously move between venues (every 30-100 cycles)
- Interactions occur at 20%-50% frequency based on sociability
- FIPA messages appear every ~50 cycles
- Global happiness fluctuates based on compatibility matches
- Analysis printed every 500 cycles